

Note 1.3 Propositional Equivalences

Introduction

- This section introduces the methods that produce propositions with the same truth value.
- Definition of tautology, contradiction and contingency**
A tautology is a compound proposition that is always true.
A contradiction is a compound proposition that is always false.
A contingency is a compound proposition that is neither a tautology nor a contradiction.

Logical Equivalences

- Definition of Logically Equivalent:**
The compound propositions p and q are called logically equivalent if $p \leftrightarrow q$ is a tautology, denoted by $p \equiv q$.
Note that $\equiv$ is not a logical connective and $p \equiv q$ is not a compound proposition but a statement that $p \leftrightarrow q$ is a tautology. Sometimes $\Leftrightarrow$ is used instead of $\equiv$.
- Table for Logical Equivalences:

Table 6	Logical Equivalences
Equivalence	Name
$p \wedge \text{T} \equiv p, p \vee \text{F} \equiv p$	Identity laws
$p \vee \text{T} \equiv \text{T}, p \wedge \text{F} \equiv \text{F}$	Domination laws
$p \vee p \equiv p, p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p, p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r), (p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r), p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q, \neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p, p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \text{T}, p \wedge \neg p \equiv \text{F}$	Negation laws

Table 7 Logical Equivalences Involving Conditional Statements
$p \rightarrow q \equiv \neg p \vee q$
$p \rightarrow q \equiv \neg q \rightarrow \neg p$
$p \vee q \equiv \neg p \rightarrow q$

Table 7 Logical Equivalences Involving Conditional Statements
$p \wedge q \equiv \neg(p \rightarrow \neg q)$
$\neg(p \rightarrow q) \equiv p \wedge \neg q$
$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$
$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$
$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$
$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

Table 8 Logical Equivalences Involving Biconditional Statements
$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$
$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

- **The Extension of Associative Law and the De Morgan's Laws**

- **Associative Law:**

$p_1 \vee p_2 \vee \dots \vee p_n$ and $p_1 \wedge p_2 \wedge \dots \wedge p_n$ follows the associative laws. We sometimes use the notation $\bigvee_{j=1}^n p_j$ for $p_1 \vee p_2 \vee \dots \vee p_n$ and $\bigwedge_{j=1}^n p_j$ for $p_1 \wedge p_2 \wedge \dots \wedge p_n$.

- **De Morgan's Laws:**

$\neg(p_1 \vee p_2 \vee \dots \vee p_n) \equiv (\neg p_1 \wedge \neg p_2 \wedge \dots \wedge \neg p_n)$ and $\neg(p_1 \wedge p_2 \wedge \dots \wedge p_n) \equiv (\neg p_1 \vee \neg p_2 \vee \dots \vee \neg p_n)$ follows the De Morgan's Laws. We sometimes use the notation $\neg\left(\bigvee_{j=1}^n p_j\right) \equiv \bigvee_{j=1}^n \neg p_j$ and $\neg\left(\bigwedge_{j=1}^n p_j\right) \equiv \bigvee_{j=1}^n \neg p_j$.

Using De Morgan's Laws

- The laws tell us how to negate conjunctions and how to negate disjunctions.

Constructing New Logical Equivalences

- **Strategy:**

We use if p and q are logically equivalent and q and r are logically equivalent then p and r are logically equivalent to prove the problems.

Propositional Satisfiability

- **Definition of satisfiability:**

A compound proposition is satisfiable if there is an assignment of truth values to its variables to make the proposition true and such an assignment is called a solution. If not, it is unsatisfiable.

Applications of Satisfiability

- **Example--Sudoku Puzzle:**

- For each cell with a given value, we assert $p(i, j, n)$ when the cell in row i and column j with the value n .
- We assert that every row contains every number:

$$\bigwedge_{i=1}^9 \bigwedge_{n=1}^9 \bigvee_{j=1}^9 p(i, j, n)$$

that is, for any i from 1 to 9 and any n from 1 to 9, there exists at least one j from 1 to 9 such that the cell has the value n in row i and column j .

- We assert that every column contains every number:

$$\bigwedge_{j=1}^9 \bigwedge_{n=1}^9 \bigvee_{i=1}^9 p(i, j, n)$$

that is, for any j from 1 to 9 and any n from 1 to 9, there exists at least one i from 1 to 9 such that the cell has the value n in row i and column j .

- We assert that each of the nine 3×3 blocks contains every number:

$$\bigwedge_{r=0}^2 \bigwedge_{s=0}^2 \bigwedge_{n=1}^9 \bigvee_{i=1}^3 \bigvee_{j=1}^3 p(3r + i, 3s + j, n)$$

that is for any blocks and any n from 1 to 9, there exist at least one cell in row i and column j from 1 to 9 with the value n in the block.

- To assert that no cell contains more than one number, we take the conjunction over all values of n, n', i and j where each variable ranges from 1 to 9 and $n \neq n'$ of

$$p(i, j, n) \rightarrow \neg p(i, j, n')$$

Solving Satisfiability Problems

- **Strategy:**

Truth table, equivalently judge its negation when the number of variables is small.